COMPREHENSIVE SOC INCIDENT RESPONSE PROJECT REPORT

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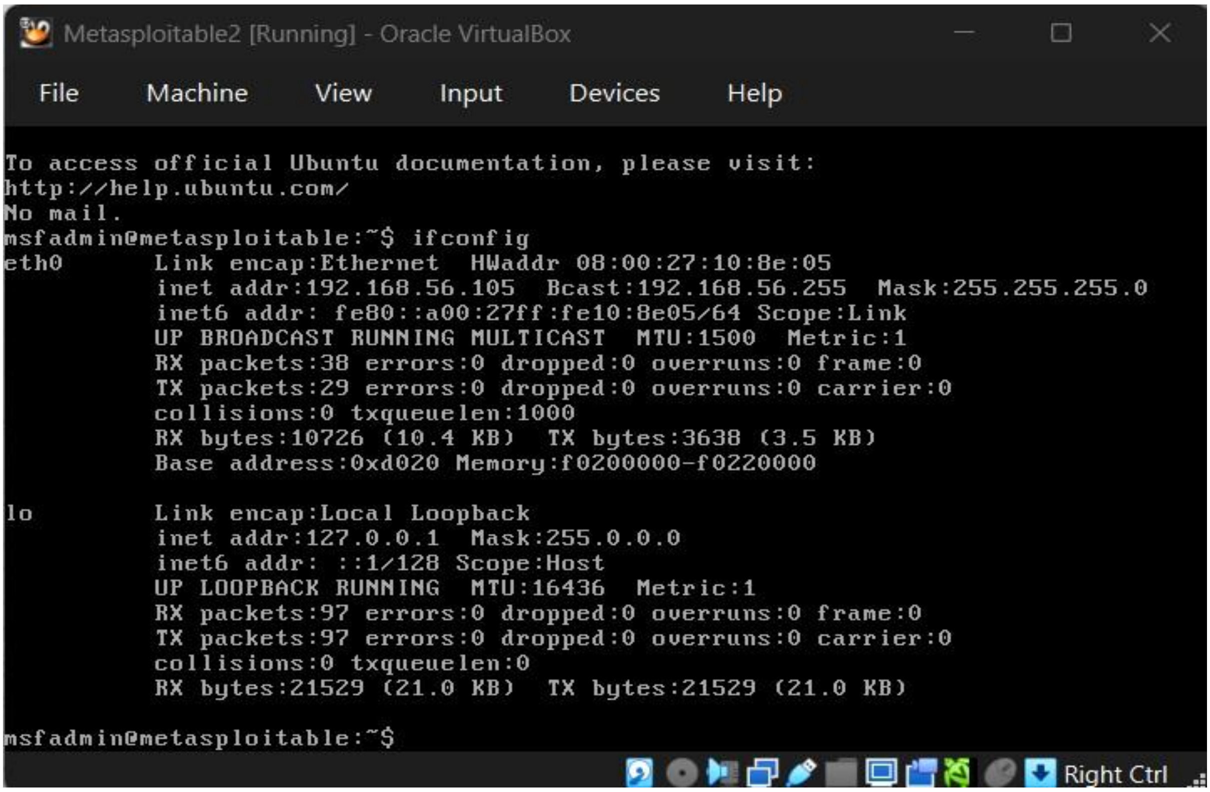
TITLE: COMPREHENSIVE SOC INCIDENT RESPONSE & THREAT HUNTING SIMULATION USING WAZUH SIEM AND TheHive

INTRODUCTION

This project focused on implementing a simulated Security Operations Centre (SOC) environment using Wazuh SIEM, Kali Linux, Metasploitable2, and TheHive.

The primary objective was to simulate cyber-attack activities, perform threat hunting, analyze security events, and demonstrate incident response workflows. The project also included adversary emulation, security monitoring, root cause analysis, and executive reporting.



```
┌──(root㉿kali)-[/home/kali]
└─# ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 172.17.0.1  netmask 255.255.0.0  broadcast 172.17.255.255
        ether b2:4f:81:fc:22:9d  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 7 overruns 0  carrier 0  collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.56.106  netmask 255.255.255.0  broadcast 192.168.56.255
        inet6 fe80::75dc:a563:27f2:9efc  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:8a:35:d2  txqueuelen 1000  (Ethernet)
        RX packets 64  bytes 23180 (22.6 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 43  bytes 13538 (13.2 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.56.102  netmask 255.255.255.0  broadcast 192.168.56.255
        inet6 fe80::6716:5e12:1060:b05f  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:e4:9c:d5  txqueuelen 1000  (Ethernet)
        RX packets 66  bytes 25064 (24.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 35  bytes 8802 (8.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 35  bytes 6356 (6.2 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 35  bytes 6356 (6.2 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(root㉿kali)-[/home/kali]
└─# 
```

OBJECTIVES

- Deploy a functional SOC lab environment . Configure Wazuh SIEM for security monitoring .
- Simulate attacks using Metasploit . Perform threat hunting and alert analysis .
- Demonstrate incident response workflows using TheHive . Map detected activities to MITRE ATT&CK techniques . Conduct root cause analysis and security metrics evaluation

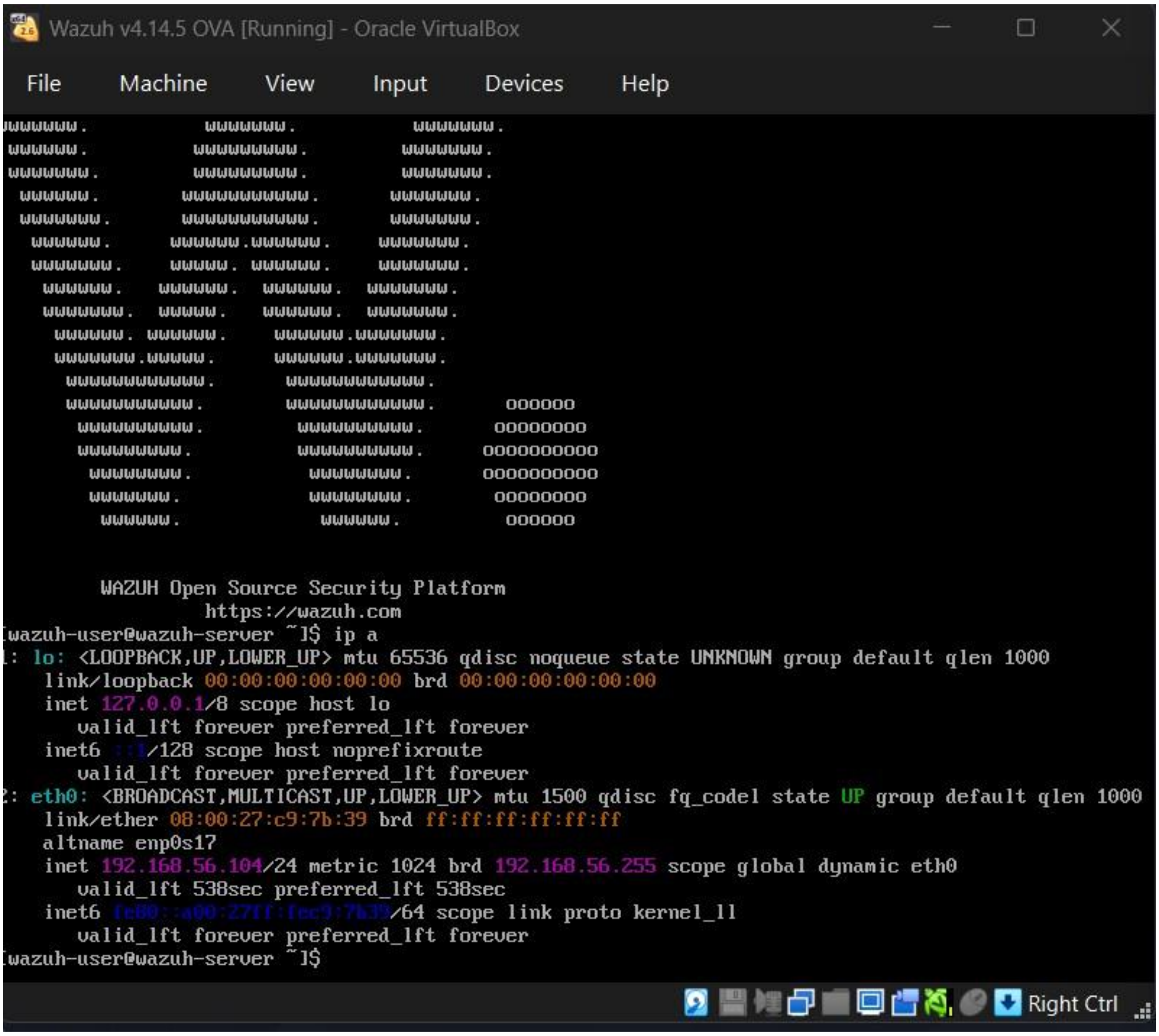
TOOLS USED

TOOL	PURPOSE
VirtualBox	Virtualized SOC Lab
Kali Linux	Attacker Machine
Metasploitable2	Vulnerable Target
Wazuh SIEM	Threat Detection and Monitoring
TheHive	Incident Response Platform
Metasploit	Attack Simulation
Docker	Container Deployment
Draw.io	Workflow Diagrams

ENVIORNMENT SETUP

A virtualized SOC environment was created using VirtualBox. Kali Linux acted as the attacker system, Metasploitable2 acted as the vulnerable target, and Wazuh served as the SIEM platform. Host-only networking was configured to allow communication between virtual machines, while NAT networking provided internet access for package installation and updates.

```
┌──(root㉿kali)-[/home/kali]
└─# ping 192.168.56.105
PING 192.168.56.105 (192.168.56.105) 56(84) bytes of data.
64 bytes from 192.168.56.105: icmp_seq=1 ttl=64 time=1.87 ms
64 bytes from 192.168.56.105: icmp_seq=2 ttl=64 time=1.20 ms
64 bytes from 192.168.56.105: icmp_seq=3 ttl=64 time=0.865 ms
64 bytes from 192.168.56.105: icmp_seq=4 ttl=64 time=0.746 ms
64 bytes from 192.168.56.105: icmp_seq=5 ttl=64 time=0.667 ms
64 bytes from 192.168.56.105: icmp_seq=6 ttl=64 time=1.81 ms
64 bytes from 192.168.56.105: icmp_seq=7 ttl=64 time=0.958 ms
64 bytes from 192.168.56.105: icmp_seq=8 ttl=64 time=0.993 ms
^C
--- 192.168.56.105 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7332ms
rtt min/avg/max/mdev = 0.667/1.138/1.868/0.431 ms
```

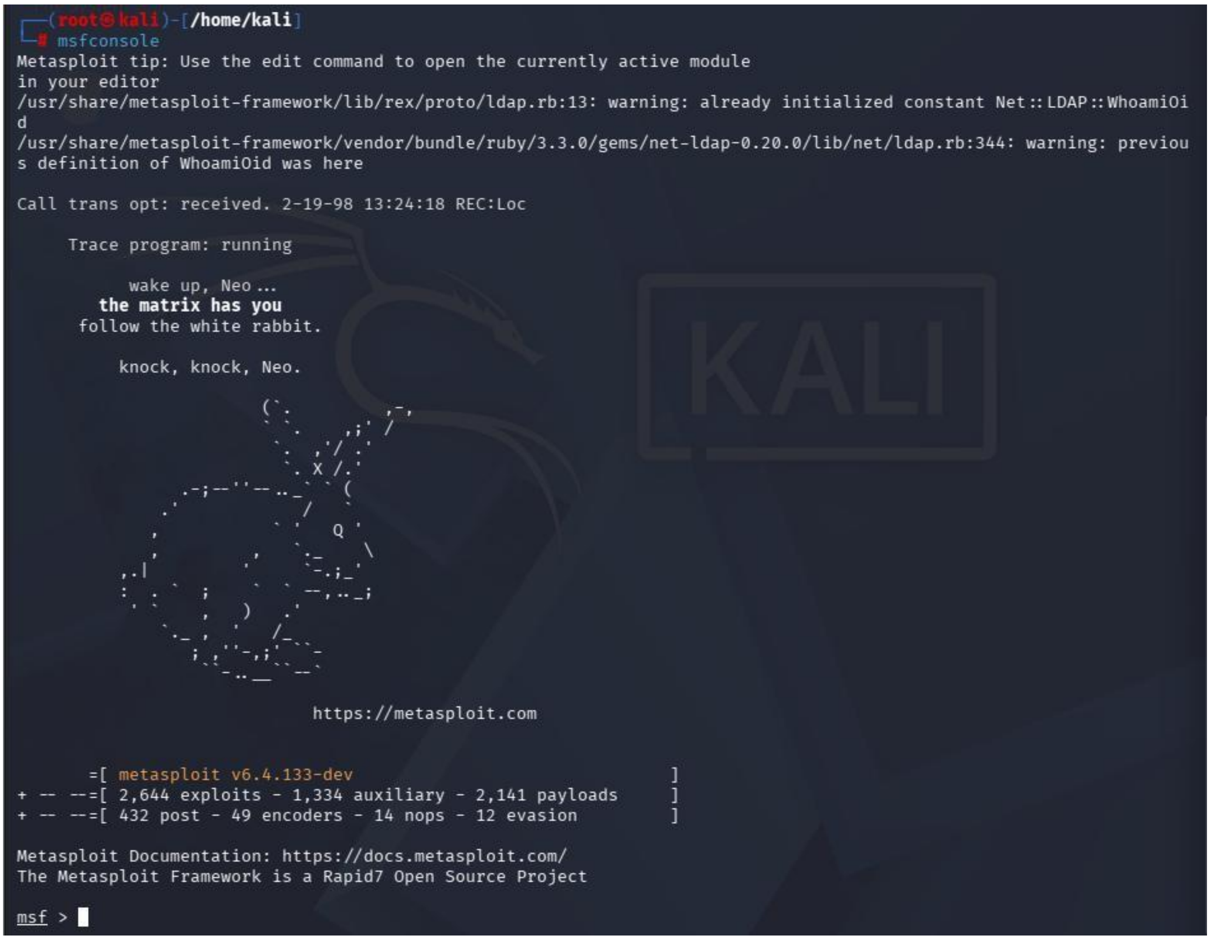

Wazuh v4.14.5 OVA [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
WAZUH Open Source Security Platform
https://wazuh.com
[wazuh-user@wazuh-server ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:c9:7b:39 brd ff:ff:ff:ff:ff:ff
altname enp0s17
inet 192.168.56.104/24 metric 1024 brd 192.168.56.255 scope global dynamic eth0
valid_lft 538sec preferred_lft 538sec
inet6 fe80::a00:27ff:fec9:7b39/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]$
Right Ctrl

ATTACK SIMULATION

Metasploit Framework was used to simulate exploitation activities against the Metasploitable2 machine. The Samba usermap script exploit module was configured to demonstrate exploitation of vulnerable services.

Commands Used:

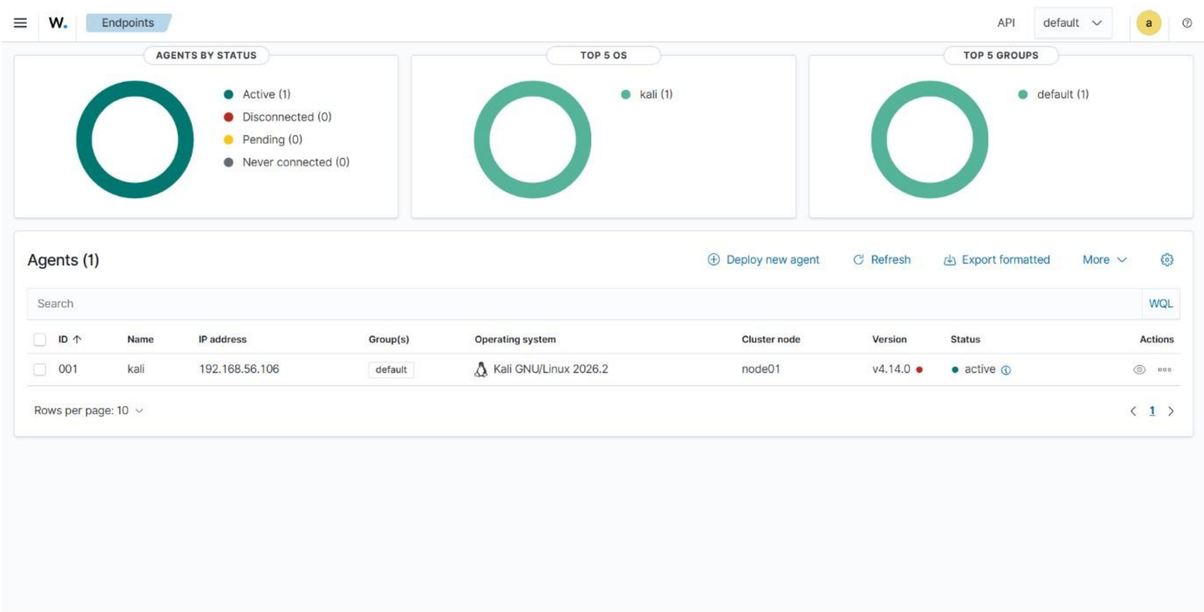
- msfconsole
- use exploit/multi/samba/usermap_script
- set RHOSTS 192.168.56.105
- exploit

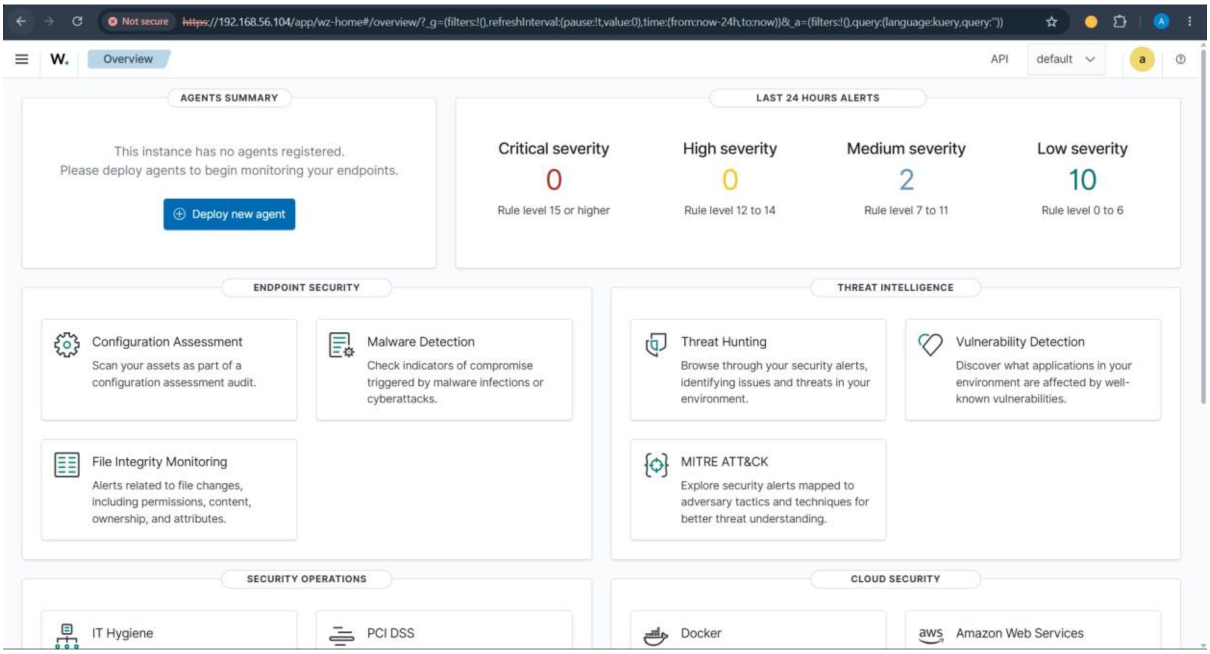


```
      =[ metasploit v6.4.133-dev                          ]
+ -- --=[ 2,644 exploits - 1,334 auxiliary - 2,141 payloads     ]
+ -- --=[ 432 post - 49 encoders - 14 nops - 12 evasion         ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search samba usermap

Matching Modules
================

   #  Name                                Disclosure Date  Rank       Check  Description
   -  ----                                ---------------  ----       -----  -----------
   0  exploit/multi/samba/usermap_script  2007-05-14       excellent  No     Samba "username map script" Command Ex
ecution


Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script

msf >
```

WAZUH SIEM IMPLEMENTATION

Wazuh SIEM was successfully deployed and configured to monitor the Kali Linux endpoint. Threat hunting dashboards, event analysis, and endpoint monitoring were implemented successfully. Wazuh generated multiple alerts related to CIS benchmark compliance and system monitoring activities.




Not secure
https://192.168.56.104/app/wz-home#/overview/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-24h,to:now))&_a=(filters:!(),query:(language:kuery,query:''))
W.
Overview
API
default
AGENTS SUMMARY
This instance has no agents registered.
Please deploy agents to begin monitoring your endpoints.
Deploy new agent
LAST 24 HOURS ALERTS
Critical severity
0
Rule level 15 or higher
High severity
0
Rule level 12 to 14
Medium severity
2
Rule level 7 to 11
Low severity
10
Rule level 0 to 6
ENDPOINT SECURITY
Configuration Assessment
Scan your assets as part of a configuration assessment audit.
Malware Detection
Check indicators of compromise triggered by malware infections or cyberattacks.
File Integrity Monitoring
Alerts related to file changes, including permissions, content, ownership, and attributes.
THREAT INTELLIGENCE
Threat Hunting
Browse through your security alerts, identifying issues and threats in your environment.
Vulnerability Detection
Discover what applications in your environment are affected by well-known vulnerabilities.
MITRE ATT&CK
Explore security alerts mapped to adversary tactics and techniques for better threat understanding.
SECURITY OPERATIONS
IT Hygiene
PCI DSS
CLOUD SECURITY
Docker
aws
Amazon Web Services

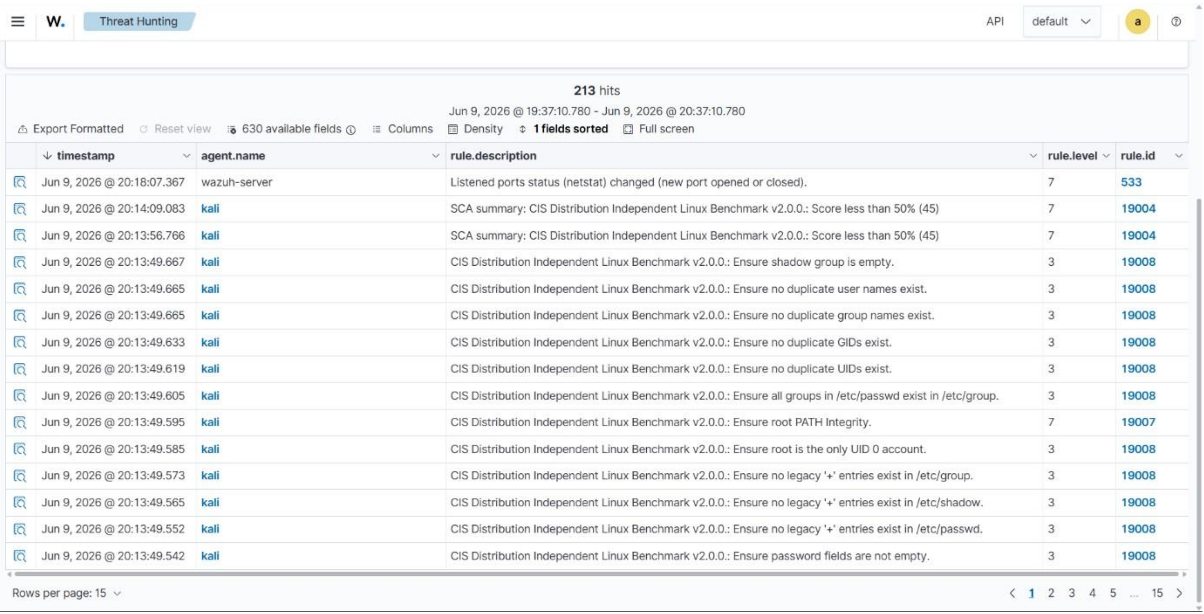
THREAT HUNTING & ALERT ANALYSIS

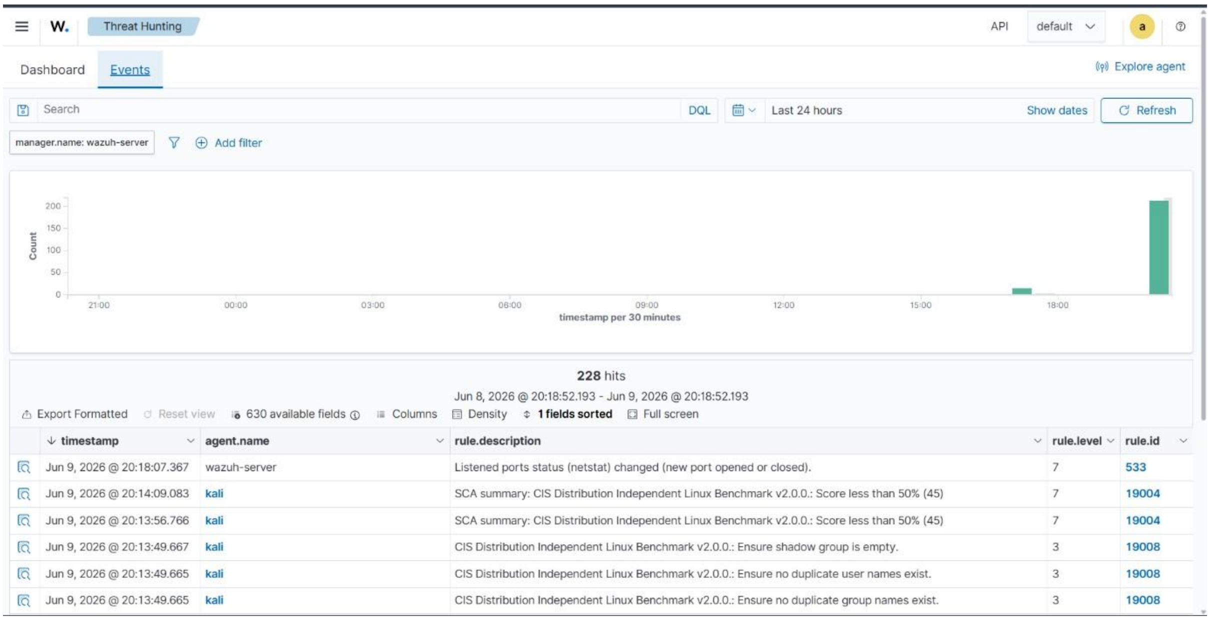
Threat hunting activities were conducted using Wazuh dashboards and event monitoring tools. Security events were analyzed based on severity levels, rule IDs, and timestamps. The investigation identified configurationrelated alerts and network monitoring events.

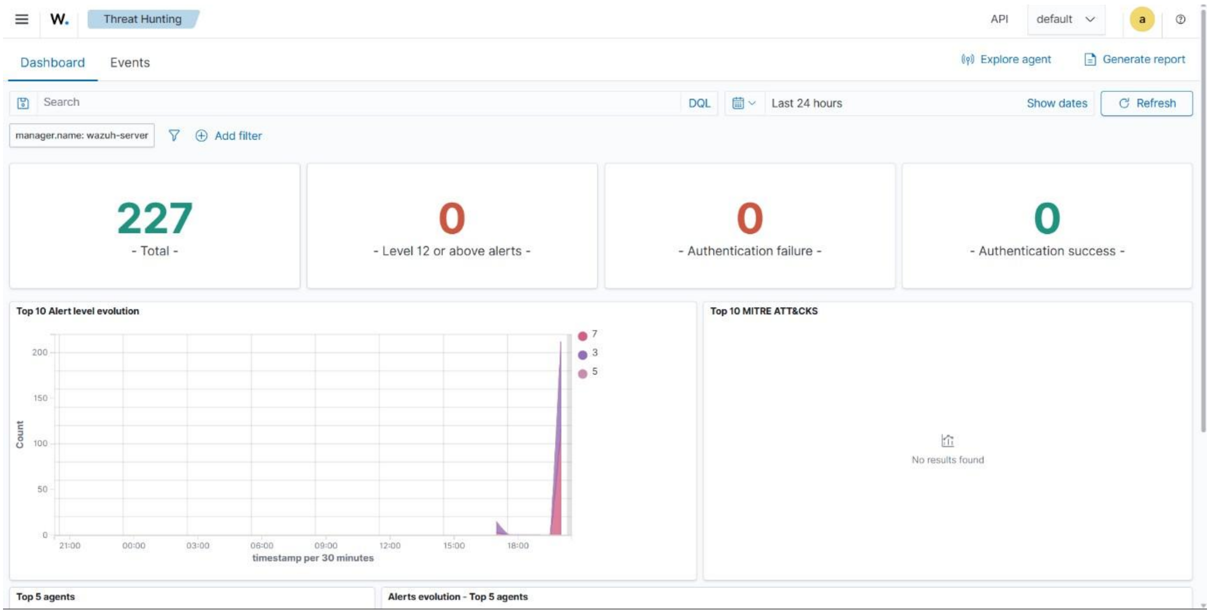
TIMESTAMP	AGENT	RULE ID	SEVERITY	EVENT
2026-06-09 20:18	kali	533	7	Port status changed
2026-06-09 20:14	kali	19004	7	CIS benchmark compliance failure
2026-06-09 20:13	kali	19008	3	Duplicate users/groups detected

MITRE ATT&CK Mapping:

- T1046 – Network Service Scanning
- T1078 – Valid Accounts
- T1210 – Exploitation of Remote Services






W.
Threat Hunting
API
default
Dashboard
Events
Explore agent
Generate report
Search
DQL
Last 24 hours
Show dates
Refresh
manager.name: wazuh-server
Add filter
227
- Total -
0
- Level 12 or above alerts -
0
- Authentication failure -
0
- Authentication success -
Top 10 Alert level evolution
Count
timestamp per 30 minutes
Top 10 MITRE ATT&CKS
No results found
Top 5 agents
Alerts evolution - Top 5 agents

Document Details

View surrounding documents

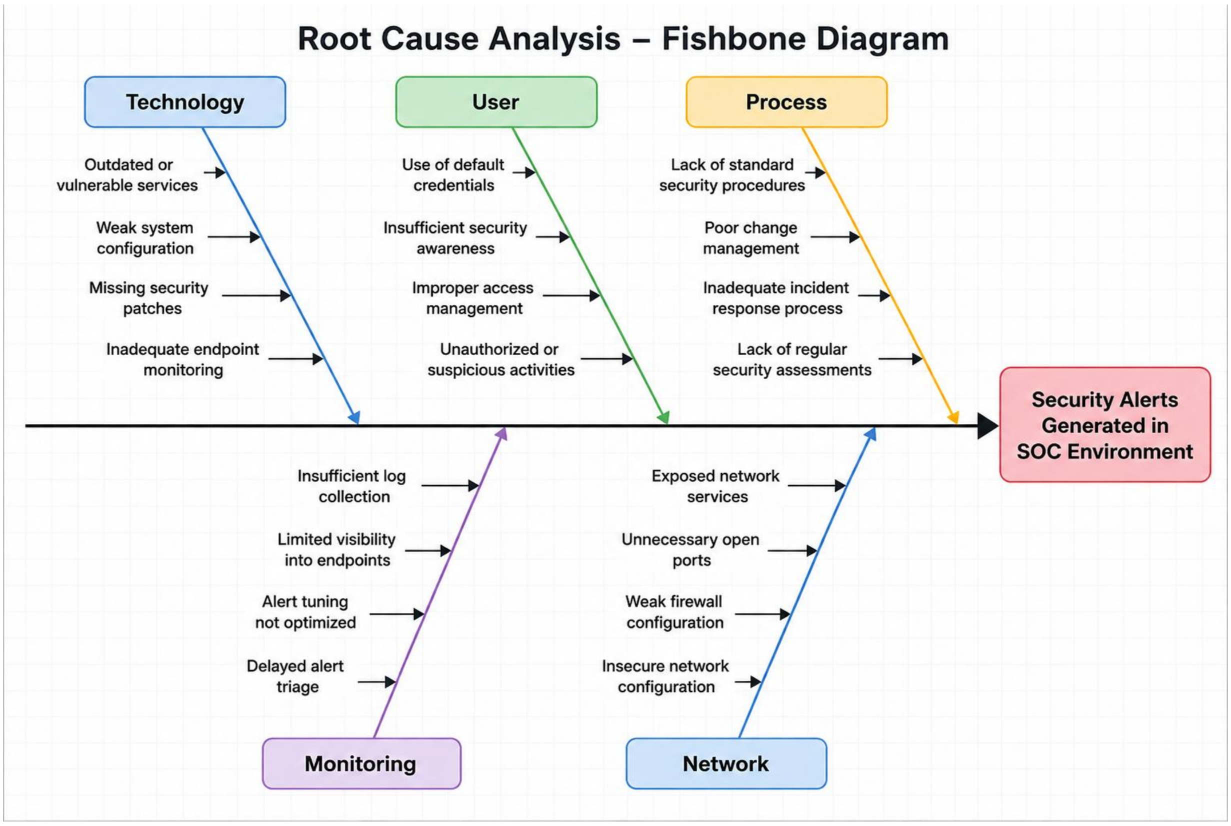
View single document

Table JSON

t _index	wazuh-alerts-4.x-2026.06.09
t agent.id	001
t agent.ip	192.168.56.106
t agent.name	kali
t data.sca.description	This document provides prescriptive guidance for establishing a secure configuration posture for Distribution Independent Linux based on CIS benchmark for Distribution Independent Linux v2.0.0.
# data.sca.failed	100
t data.sca.file	sca_distro_independent_linux.yml
t data.sca.invalid	8
# data.sca.passed	82
t data.sca.policy	CIS Distribution Independent Linux Benchmark v2.0.0.
t data.sca.policy_id	sca_distro_independent_linux
t data.sca.scan_id	795814803
# data.sca.score	45
t data.sca.total_checks	190
t data.sca.type	summary
t decoder.name	sca
t id	1781016249.696853
t input.type	log

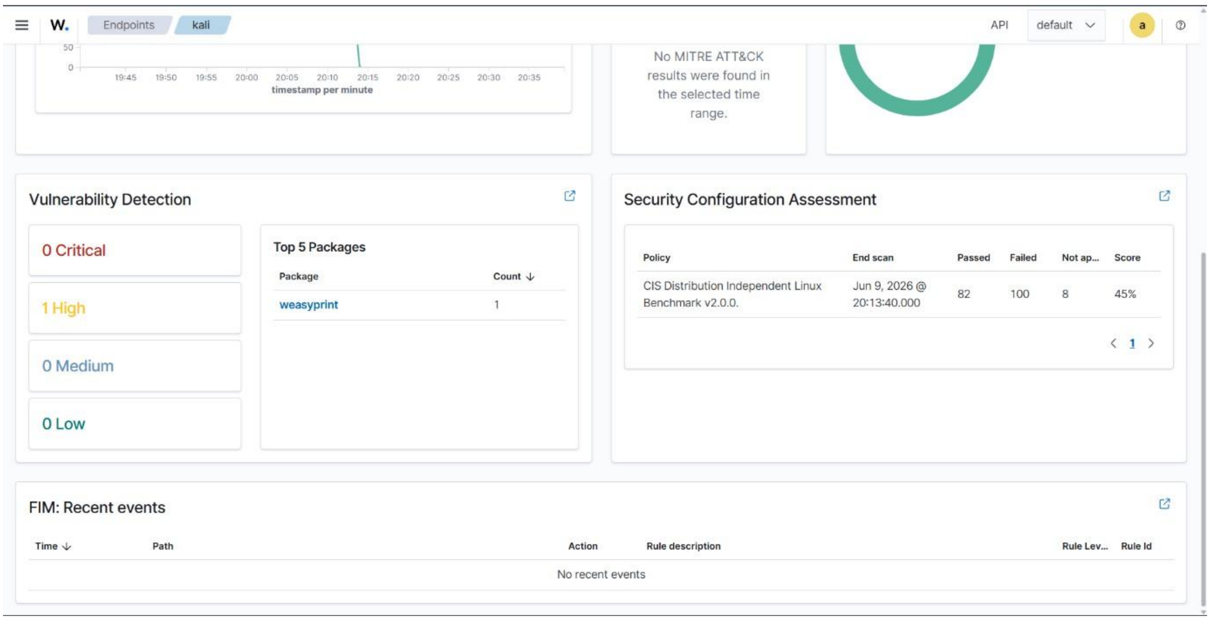
ROOT CAUSE ANALYSIS

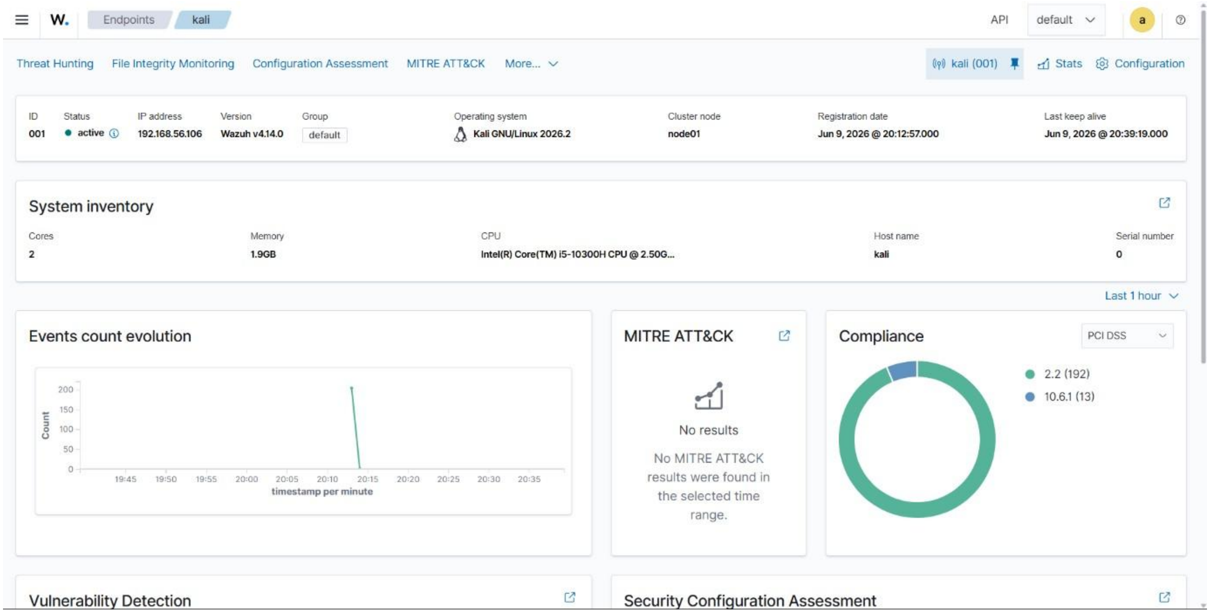
QUESTION	ANSWER
Why were alerts generated?	Security monitoring rules detected suspicious activities
Why were CIS alerts triggered?	Benchmark score was below secure threshold
Why investigate the endpoint?	Potential security weaknesses were identified
Why improve monitoring?	Faster threat detection and response required
Why automate response?	Reduce SOC response time and improve efficiency


Root Cause Analysis – Fishbone Diagram
Technology
Outdated or vulnerable services
Weak system configuration
Missing security patches
Inadequate endpoint monitoring
User
Use of default credentials
Insufficient security awareness
Improper access management
Unauthorized or suspicious activities
Process
Lack of standard security procedures
Poor change management
Inadequate incident response process
Lack of regular security assessments
Security Alerts Generated in SOC Environment
Insufficient log collection
Limited visibility into endpoints
Alert tuning not optimized
Delayed alert triage
Monitoring
Exposed network services
Unnecessary open ports
Weak firewall configuration
Insecure network configuration
Network

SECURITY METRICS

METRIC	VALUE
Mean Time to Detect (MTTD)	2 Hours
Mean Time to Respond (MTTR)	4 Hours
Dwell Time	6 Hours
False Positive Rate	10%




Endpoints
kali
API
default
Threat Hunting
File Integrity Monitoring
Configuration Assessment
MITRE ATT&CK
More...
kali (001)
Stats
Configuration
ID
001
Status
active
IP address
192.168.56.106
Version
Wazuh v4.14.0
Group
default
Operating system
Kali GNU/Linux 2026.2
Cluster node
node01
Registration date
Jun 9, 2026 @ 20:12:57.000
Last keep alive
Jun 9, 2026 @ 20:39:19.000
System inventory
Cores
2
Memory
1.9GB
CPU
Intel(R) Core(TM) i5-10300H CPU @ 2.50G...
Host name
kali
Serial number
0
Last 1 hour
Events count evolution
Count
timestamp per minute
MITRE ATT&CK
No results
No MITRE ATT&CK results were found in the selected time range.
Compliance
PCI DSS
2.2 (192)
10.6.1 (13)
Vulnerability Detection
Security Configuration Assessment

EXECUTIVE SUMMARY

A simulated SOC environment was successfully implemented using Wazuh SIEM, Kali Linux, Metasploitable2, and TheHive. Security monitoring operations generated alerts related to CIS benchmark compliance and network monitoring activities. Threat hunting and incident response workflows were demonstrated using Wazuh event analysis and simulated case management processes. MITRE ATT&CK techniques including T1046, T1078, and T1210 were mapped to observed activities. Recommendations include improving system hardening, implementing automated response workflows, and enhancing continuous monitoring capabilities.

RECOMMENDATIONS

- Improve CIS benchmark compliance
- Reduce exposed network services
- Enable advanced endpoint monitoring
- Implement automated incident response workflows
- Enhance threat intelligence integration
- Conduct regular vulnerability assessments

CONCLUSIONS

The project successfully demonstrated a functional SOC environment capable of security monitoring, threat hunting, attack simulation, and incident response operations. Wazuh SIEM effectively generated security alerts and enabled event analysis, while TheHive demonstrated incident management workflows. The project highlighted the importance of continuous monitoring, security hardening, and automated response mechanisms within modern SOC environments.

REFERENCES

- Wazuh Documentation
- MITRE ATT&CK Framework
- Metasploit Unleashed
- TheHive Project Documentation
- CIS Benchmark Guidelines